DATA ANALYTICS PROJECT

PIZZA SALES

PERFORMANCE ANALYSIS REPORT

January – December 2015

$817,860	21,350	49,574	32
Total Revenue	Total Orders	Pizzas Sold	Pizza Varieties

Tool Used	Dataset	Prepared By
Microsoft Excel 2021	**48,620 rows • 14 columns**	**Aman Singh Rawat**

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1. Project Overview & Objectives

This report presents a complete data analytics project built entirely in Microsoft Excel, analyzing pizza sales data for the full calendar year 2015. The project was designed to simulate a real-world business intelligence scenario where a pizza restaurant needs to understand its sales patterns, identify best-performing products, optimize operations, and communicate findings through a professional Excel dashboard.

The entire project — from raw data storage and cleaning, through Pivot Table analysis, formula-based KPI calculation, chart creation, and final dashboard design — was completed using only Microsoft Excel. No external tools were used.

1.1 Project Objectives

- Calculate and monitor five core KPIs: Total Revenue, Total Orders, Total Pizzas Sold, Average Order Value, and Average Pizzas per Order.
- Analyse which pizza categories (Classic, Supreme, Chicken, Veggie) generate the highest revenue and order volume using Pivot Tables.
- Understand how pizza size (Small, Medium, Large, X-Large, XX-Large) impacts revenue distribution.
- Identify the Top 5 best-selling pizzas and Bottom 5 worst-selling pizzas by quantity to guide menu decisions.
- Discover peak hours and peak days for orders to support staffing and operational planning.
- Track monthly revenue trends across all 12 months of 2015 to identify seasonal patterns.
- Build an interactive Excel Dashboard combining KPI cards, pivot charts, and formatted summaries for stakeholder reporting.

1.2 Tools & Features Used

Excel Feature	How It Was Used in This Project
Pivot Tables	Aggregated revenue, quantity, and order counts by category, size, day, and pizza name
Pivot Charts	Bar charts, donut/pie charts, and line charts linked to pivot tables for dynamic visualization
Power Query (Get & Transform)	Data cleaning pipeline — type conversions, derived columns (Month, Hour), text standardization
Excel Formulas	SUM, COUNT, AVERAGE, SUMIF, COUNTIF, TEXT, MONTH, HOUR for KPI calculations
Conditional Formatting	Highlighted top/bottom performers with color scales and data bars in summary tables
Dashboard Sheet	Dedicated dashboard tab combining all KPI boxes, charts, and insights in a single professional view
Slicers	Connected to pivot tables for interactive filtering by pizza category and size

1.3 Project Workflow

Step 1	Step 2	Step 3	Step 4
Data Cleaning	Pivot Analysis	KPI Formulas	Dashboard
Power Query	*Pivot Tables*	*Excel Formulas*	*Excel Charts*

2. Dataset Description

The dataset is a transactional pizza sales file containing 48,620 individual pizza order line items for the year 2015. Each row represents one type of pizza ordered within a specific customer order, with detailed information about the order, the pizza, pricing, timing, and category. The data was stored in the pizza_sales sheet of the Excel workbook as the primary raw data source.

2.1 Workbook Structure

Sheet Name	Type	Purpose
pizza_sales	Raw Data	Primary dataset — 48,620 rows, 14 columns of transactional data
KPI	Summary	5 key metrics: Total Revenue, Orders, Pizzas Sold, Avg Order Value, Avg Pizzas/Order
Trends for total orders	Pivot	Daily (day-of-week) and Hourly pivot tables showing order volume patterns
Percentage of sales	Pivot	Revenue % by category, revenue % by size, and quantity by category
Best and worst seller	Pivot	Top 5 and Bottom 5 pizzas ranked by quantity sold
Dashboard	Visual	Final Excel dashboard combining KPI cards, all charts, and business insights

2.2 Data Dictionary — pizza_sales Sheet

Column Name	Data Type	Description
pizza_id	Integer	Unique row identifier for each pizza line item
order_id	Integer	Customer order number — multiple rows share same ID for multi-item orders
total_orders	Float	Aggregation support column for order counting in pivot tables
pizza_name_id	String	Short code ID for the pizza (e.g., 'ckn_arto', 'bbq_ckn')
quantity	Integer	Number of this specific pizza ordered in the line item
order_date	Date	Date the order was placed (converted from text using Power Query)
order_day	String	Day of the week (Monday–Sunday) — text field for grouping
order_time	Time	Time the order was placed — used for hourly trend analysis

unit_price	Decimal	Price of a single pizza in USD
total_price	Decimal	Line item total = unit_price × quantity
pizza_size	String	Size of pizza: S (Small), M (Medium), L (Large), XL (X-Large), XXL (XX-Large)
pizza_category	String	Category: Classic, Supreme, Chicken, or Veggie
pizza_ingredients	String	Comma-separated list of all toppings and ingredients
pizza_name	String	Full display name (e.g., 'The Classic Deluxe Pizza')

3. Data Cleaning & Preparation

Before any analysis could be done, the raw pizza_sales data required systematic cleaning and preparation. This was accomplished entirely within Excel using Power Query (Get & Transform Data) — Excel's built-in ETL engine. Every transformation was recorded as a reproducible step in the Applied Steps panel, ensuring full auditability.

3.1 Opening Power Query in Excel

To open Power Query: Data tab → Get Data → From File → From Workbook → select pizza_sales_excel_file.xlsx → select pizza_sales table → Transform Data. This loads the raw data into the Power Query Editor where all cleaning steps can be applied.

3.2 Issues Found in Raw Data

- The order_date column was imported as a Text type instead of Date — this prevented proper time-based filtering and sorting in Pivot Tables.
- The order_time column was text-formatted, making hourly analysis impossible without type conversion.
- The unit_price and total_price columns were imported as 'Any' type with potential decimal inconsistencies.
- No Month Name column existed — required for monthly trend Pivot Charts.
- No Hour column existed — required for hourly traffic analysis.
- The pizza_size column used abbreviated codes (S, M, L, XL, XXL) which needed mapping to full names for clarity in charts.

3.3 Cleaning Steps Applied in Power Query

Step	Action in Power Query	Business Impact
1	**Changed order_date type → Date**	Enables MONTH(), YEAR() functions and date-based Pivot Table grouping
2	**Changed order_time type → Time**	Enables HOUR() extraction for hourly order trend analysis
3	**Changed unit_price → Decimal Number**	Ensures currency precision in all revenue calculations
4	**Changed total_price → Decimal Number**	Eliminates floating point errors in SUM formulas and Pivot aggregations

5	**Changed quantity → Whole Number**	Ensures integer counts for quantity-based pivot tables
6	**Added Column: Month Name**	Used TEXT(order_date, 'MMMM') — enables monthly grouping in pivot charts
7	**Added Column: Month Number**	Used MONTH(order_date) — ensures Jan-Dec chronological sorting in charts
8	**Added Column: Hour**	Used HOUR(order_time) — powers the hourly order trend chart on dashboard
9	**Trimmed all text columns**	Removes invisible leading/trailing spaces preventing mis-grouping in Pivot Tables
10	**Replaced size codes with full names**	S→Small, M→Medium, L→Large, XL→X-Large, XXL→XX-Large for readable chart labels

3.4 Pivot Tables Created

After loading the cleaned data into Excel, the following Pivot Tables were created on dedicated sheets to power the dashboard analyses:

Pivot Table Name / Sheet	Rows / Values / Analysis
Trends for total orders	Row: order_day | Row: Hour | Value: Sum of total_orders — for daily and hourly trends
Percentage of sales (Category)	Row: pizza_category | Value: Sum of total_price as % of Grand Total
Percentage of sales (Size)	Row: pizza_size | Value: Sum of total_price as % of Grand Total
Percentage of sales (Qty by Cat)	Row: pizza_category | Value: Sum of quantity — for category quantity comparison
Best and worst seller	Row: pizza_name | Value: Sum of quantity | Top 5 and Bottom 5 filtered views
KPI Sheet	Formula-based cells: SUMIF, COUNTIF, AVERAGE for 5 key metrics

3.5 Before vs. After Cleaning

Before Cleaning	After Cleaning
Dates stored as Text — not sortable	Proper Date type — full Pivot Table date grouping works
No Month Name column	Month Name added — monthly charts display correctly
No Hour column	Hour extracted — hourly trend chart enabled
Size codes: S, M, L, XL, XXL	Full names: Small, Medium, Large, X-Large, XX-Large
14 original columns	17 columns after 3 new derived columns added

4. Key Performance Indicators (KPIs)

Five core KPIs were calculated on the dedicated KPI sheet using Excel SUM, COUNT, and AVERAGE formulas referencing the cleaned pizza_sales table. These KPIs were then displayed as formatted KPI cards on the final Excel dashboard.

$817,860.05 **TOTAL REVENUE** `=SUM(pizza_sales[total_price])`	**21,350** **TOTAL ORDERS** `=SUMPRODUCT(1/COUNTIF(order_id,order_id))`
49,574 **TOTAL PIZZAS SOLD** `=SUM(pizza_sales[quantity])`	**$38.31** **AVG ORDER VALUE** `=Total_Revenue / Total_Orders`

2.32 Pizzas per Order

AVERAGE PIZZAS PER ORDER | Formula: =Total_Pizzas_Sold / Total_Orders

KPI Interpretation: With $817,860 revenue from 21,350 orders across the full year, the business averages approximately $2,241 in daily revenue and 58 orders per day. An average of 2.32 pizzas per order strongly indicates group dining behavior, suggesting most customers order for families or small groups rather than individual meals.

5. Sales Analysis by Pizza Category

The Pivot Table on the 'Percentage of sales' sheet was used to aggregate revenue and quantity by the four pizza categories. This analysis answers the core business question: which category is driving the most revenue and volume?

5.1 Revenue & Quantity by Category — Pivot Table Results

Pizza Category	Revenue	Revenue %	Qty Sold	Orders
Classic	**$220,053**	**26.91%**	14,888	10,859
Supreme	$208,197	25.46%	11,987	9,085
Chicken	$195,920	23.96%	11,050	8,536
Veggie	$193,690	23.68%	11,649	8,941
TOTAL	**$817,860**	**100%**	**49,574**	—

5.2 Insights

- Classic pizzas lead in both revenue ($220,053 / 26.91%) and quantity sold (14,888 units) — confirming that traditional, familiar flavors have the strongest appeal among customers.
- Supreme and Chicken categories are close competitors (25.46% and 23.96%), showing healthy demand for premium and specialty options.
- Veggie category contributes a competitive 23.68% revenue share despite being a niche segment — indicating a meaningful health-conscious customer base.
- The near-equal distribution across all four categories (23.68%–26.91%) shows the menu is well-balanced, with no single category critically over-relied upon.

6. Sales Analysis by Pizza Size

The size-level analysis from the Percentage of Sales Pivot Table reveals a clear customer preference pattern that has direct implications for dough preparation, pricing, and portion strategy.

6.1 Revenue & Quantity by Size

Size	Revenue	Revenue %	Qty Sold	Qty %
Large	**$375,319**	**45.89%**	18,956	38.24%
Medium	$249,382	30.49%	15,635	31.54%
Small	$178,077	21.77%	14,403	29.05%
X-Large	$14,076	1.72%	552	1.11%
XX-Large	$1,007	0.12%	28	0.06%

6.2 Insights

- Large pizzas are overwhelmingly preferred — 45.89% of revenue and 38.24% of quantity. Customers consistently upsize, likely driven by value perception and group sharing behavior.
- Large + Medium together account for 76.38% of all revenue — these two sizes are the backbone of the business.
- Small pizzas represent 29% of quantity but only 21.77% of revenue, indicating lower per-unit price points for individual or lunch diners.
- X-Large (1.72%) and XX-Large (0.12%) together contribute under 2% of revenue — suggesting these sizes may be unnecessary menu complexity.

7. Best & Worst Selling Pizzas

The 'Best and worst seller' sheet contains two Pivot Tables — one filtered to the Top 5 pizzas and one for the Bottom 5, both ranked by Sum of Quantity. These directly power the horizontal bar chart visible on the Excel dashboard.

7.1 Top 5 Best Sellers — by Quantity

#	Pizza Name	Qty Sold	Revenue
1	The Classic Deluxe Pizza	**2,453**	**$38,181**
2	The Barbecue Chicken Pizza	2,432	$42,768
3	The Hawaiian Pizza	2,422	$32,273
4	The Pepperoni Pizza	2,418	$30,162
5	The Thai Chicken Pizza	2,371	$43,434

7.2 Bottom 5 Worst Sellers — by Quantity

#	Pizza Name	Qty Sold	Revenue
32	**The Brie Carre Pizza**	**490**	$11,589
31	The Mediterranean Pizza	934	$15,361
30	The Calabrese Pizza	937	$15,934
29	The Spinach Supreme Pizza	950	$15,278
28	The Soppressata Pizza	961	$16,426

The Brie Carre Pizza sold only 490 units — just 20% of the #1 seller's volume. With a revenue of $11,589 it is the lowest-earning pizza on the menu and the primary candidate for removal or reformulation to simplify operations and reduce ingredient waste.

8. Hourly & Daily Order Trends

The 'Trends for total orders' sheet contains two separate Pivot Tables: one showing orders by Day of Week and one showing orders by Hour of Day. These were used to create the bar charts on the Excel dashboard.

8.1 Orders by Day of Week

Rank	Day	Total Orders	Revenue	vs Sunday
1	**Friday (Peak Day)**	**3,538**	**$136,074**	**+34.8%**
2	**Thursday**	3,239	$123,529	+23.4%
3	**Saturday**	3,158	$123,182	+20.3%
4	**Wednesday**	3,024	$114,408	+15.2%
5	**Tuesday**	2,973	$114,134	+13.3%
6	**Monday**	2,794	$107,330	+6.5%
7	**Sunday (Lowest)**	2,624	$99,204	Baseline

8.2 Peak Hours — Orders by Hour of Day

Hour	Time Period	Total Orders	Category
12:00	**Lunch Peak**	**2,520**	**Peak**
13:00	Post-Lunch	2,455	High
18:00	Dinner Start	2,399	High
17:00	Early Evening	2,336	High
19:00	Peak Dinner	2,009	Medium
09–10	Morning (Very Low)	9	Minimal

9. Monthly Revenue Trend

Monthly revenue was calculated by grouping the Pivot Table on order_date by Month. The resulting month-by-month breakdown reveals clear seasonal patterns across the 12 months of 2015.

9.1 Month-by-Month Revenue Table

#	Month	Revenue	Orders	Qty Sold	vs Monthly Avg
1	**January**	$69,793	1,845	4,232	Above Average
2	**February**	$65,160	1,685	3,961	Below Average
3	**March**	$70,397	1,840	4,261	Above Average
4	**April**	$68,737	1,799	4,151	Average
5	**May**	$71,403	1,853	4,328	Above Average
6	**June**	$68,230	1,773	4,107	Average
7	**July**	$72,558	1,935	4,392	**PEAK MONTH**
8	**August**	$68,278	1,841	4,168	Average
9	**September**	$64,180	1,661	3,890	Below Average
10	**October**	$64,028	1,646	3,883	Below Average
11	**November**	$70,395	1,792	4,266	Above Average
12	**December**	$64,701	1,680	3,935	Below Average

Monthly Average: $68,155 per month | July is the highest at $72,558 | October is the lowest at $64,028

Insight: Summer months (May, July) consistently outperform the monthly average, while Q4 (September–December) shows a declining revenue trend. November breaks this Q4 pattern with $70,395, possibly driven by holiday gatherings. January shows strong performance likely due to New Year celebrations and group outings.

10. Excel Dashboard — Charts & Pivot Tables

The final phase of the project involved building a comprehensive, single-sheet Excel Dashboard by assembling all KPI metrics, pivot charts, and business insights into a professionally formatted Excel sheet. The dashboard was designed for non-technical stakeholders — providing instant visual answers to all key business questions without requiring any interaction with the raw data.

10.1 Dashboard Layout

- Top Banner: Pizza image, project title (PIZZA SALE), and the two headline KPIs — Total Revenue ($8,17,860 displayed) and Average Order Value ($38.31).
- Left Column — Insights Panel: Text-based business insight boxes for Days, Times, Category, Size, and Best/Worst sellers — formatted with colored headings for quick scanning.
- Right Column — Chart Area: Contains all visual charts linked to Pivot Tables for dynamic updating.

10.2 Charts Created in Excel

Chart Type	Data Source	Business Question Answered
Bar Chart (Horizontal)	Daily Orders Pivot	Which day has the most/least orders? → Friday peak, Sunday lowest
Donut / Pie Chart	Category % Pivot	Which pizza category generates the most revenue? → Classic leads at 26.91%
Bar Chart (Horizontal)	Best Seller Pivot	Which pizzas sell the most? → Top 5 and Bottom 5 in one chart
KPI Cards (Text Boxes)	KPI Sheet Formulas	Revenue, Orders, Avg Order Value — displayed in formatted Excel shapes
Insight Text Boxes	Manual Summary	Written analysis of Days trends, Category findings, Size patterns, and Seller performance

10.3 Dashboard Design Principles Applied

- Dark background theme (dark brown/black) with red and gold accents — matching the pizza brand aesthetic and creating high visual contrast.
- KPIs prominently placed in the header area with large bold font for instant executive-level reading.
- Insight text boxes with colored headings (Days, Times, Category, Size, Best/Worst) for quick visual navigation.
- Charts linked to Pivot Tables so the entire dashboard automatically refreshes when source data is updated.
- Minimal gridlines and borders — clean, publication-ready layout without visual noise.

11. Key Findings & Business Insights

The following summarizes the most actionable business conclusions drawn from the full Excel analysis:

1. Revenue Performance

$817,860 generated from 21,350 orders in 2015. Daily average of ~$2,241 revenue and ~58 orders confirms a well-established, consistently busy operation.

2. Category Balance

All four categories contribute remarkably equally (23.68%–26.91% revenue share). Classic leads but no category dominates — a healthy, balanced menu portfolio.

3. Size Strategy

Large pizzas account for 45.89% of all revenue — nearly half the business. Large + Medium = 76.38% combined, making these two the critical sizes for business sustainability.

4. Top Seller — Classic Deluxe

The Classic Deluxe Pizza is the highest-volume seller at 2,453 units, confirming strong customer loyalty to traditional, straightforward toppings.

5. Worst Seller — Brie Carre

At only 490 units and $11,589 revenue, The Brie Carre Pizza is significantly underperforming. It sold 80% fewer units than the top seller and is the clearest menu optimization opportunity.

6. Peak Day — Friday

Friday generates 3,538 orders and $136,074 revenue — 34.8% more orders than Sunday. The Thu-Fri-Sat cluster is the highest-traffic period requiring maximum operational readiness.

7. Peak Hours — Lunchtime

12:00–13:00 is the busiest order window (2,520 and 2,455 orders respectively), followed by the dinner period (17:00–18:00). Morning hours (09:00–10:00) are nearly dead with only 9 total orders.

8. Peak Month — July

July peaks at $72,558 and 1,935 orders — 6.4% above the monthly average. The Q4 slowdown (Sep–Dec) presents an opportunity for seasonal promotions.

12. Recommendations

#	Focus Area	Recommendation
1	**Remove Brie Carre Pizza**	With only 490 units sold, The Brie Carre Pizza cannot justify its presence on a 32-item menu. Remove it or replace with a new flavor to simplify kitchen operations.
2	**Friday Staffing & Prep**	Maximize dough preparation and staff allocation every Friday. Consider a Friday Special promotion to push average order value above $40 during the peak traffic period.
3	**Sunday Deal Campaign**	Launch a Sunday Family Bundle (e.g., 2 Large pizzas + drinks at a discount) to lift Sunday's 2,624 orders closer to the Friday benchmark of 3,538.
4	**July Summer Campaign**	Build on July's natural peak with a summer limited-edition pizza or themed promotion to push revenue beyond the $72,558 peak.

5	**Q4 Revival Strategy**	Address the September–December revenue decline with a Diwali/Festive Season deal in October and a Christmas Family Pack in December.
6	**X-Large Upsell Initiative**	X-Large accounts for only 1.72% of revenue. Introduce an 'X-Large Party Pack' combo (pizza + garlic bread + drinks) to incentivize upsizing and increase per-order revenue.
7	**Classic Category Priority**	Classic is the revenue and volume leader — feature Classic bestsellers (Classic Deluxe, Hawaiian, Pepperoni) prominently in social media marketing and loyalty reward programs.
8	**Lunch Hour Efficiency**	12:00–13:00 is the peak lunch window. Ensure pre-made dough, pre-chopped toppings, and maximum oven capacity are ready by 11:30 to minimize wait times during lunch rush.

13. Conclusion

This project successfully delivered a complete, professional-grade data analytics solution using Microsoft Excel as the sole tool. Starting with 48,620 rows of raw transactional pizza sales data, the project moved through a disciplined four-step process: data cleaning via Power Query, KPI calculation through Excel formulas, business analysis through Pivot Tables, and visual storytelling through an Excel Dashboard.

The analysis surfaced clear, actionable insights: Friday and July are the peak trading periods requiring operational readiness; Large pizzas dominate at 45.89% of revenue; Classic category leads overall; The Brie Carre Pizza is a significant underperformer warranting menu review; and the lunch window (12:00–13:00) is the busiest order period of the day.

From a technical standpoint, this project demonstrates proficiency in the full Microsoft Excel analytics stack — Power Query for ETL, Pivot Tables for data aggregation, Excel formulas for calculated KPIs, Pivot Charts for visualization, and Dashboard design for stakeholder communication. These are exactly the skills expected of a professional data analyst in business settings.

13.1 Skills Demonstrated

Skill	Applied In This Project
Power Query / ETL	Type conversions, derived columns (Month, Hour), text trimming and standardization
Excel Pivot Tables	Category, size, daily, hourly, and pizza-name level aggregations with filtering
Excel Formulas	SUM, COUNTIF, AVERAGE, SUMIF for KPI sheet; TEXT, MONTH, HOUR for derived columns
Pivot Charts	Bar charts and donut charts linked to pivot tables for interactive, auto-updating visuals
Dashboard Design	Professional Excel dashboard with KPI cards, charts, insight panels, and brand-consistent styling
Data Storytelling	Translating raw numbers into clear business insights, trends, and actionable recommendations

"Data in Excel is not just numbers in cells —
it is the complete story of a business, waiting to be told."

14. Excel Dashboard

The following page shows the actual Excel Dashboard created for this project. The dashboard was built on a dedicated 'Dashboard' sheet in the workbook, combining KPI headline figures, a daily orders bar chart, a pizza category donut chart, and a best/worst seller horizontal bar chart — all designed with a dark, professional theme.

Figure 1: Excel Dashboard — Pizza Sales Analysis 2015